

Yoon-Joo Kim

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Education

- 2016–2022 **Ph.D. Mathematics**, Stony Brook University.
Advisor: Radu Laza.
Thesis: Constructing dual Lagrangian fibrations of compact hyper-Kähler manifolds.
- 2008–2015 B.S. Mathematics, Seoul National University, Korea.
- 2008–2015 B.S. Computer Science, Seoul National University, Korea.

Positions

- 2023–now **Ritt Assistant Professor**, Columbia University.
- 2024–2026 **June Huh visiting fellow**, Korea Institute for Advanced Study (KIAS).
- 2022–2023 **Postdoctoral researcher**, ERC HyperK team, Universität Bonn.
Project leader: D. Huybrechts (with O. Debarre, E. Macrì, and C. Voisin).

Papers and Preprints

1. *Obstructions for codimension 1 multiple fibers of Lagrangian and Calabi–Yau fibrations*,
(with K. Oguiso and E. Shinder), [arXiv:2608.10102](https://arxiv.org/abs/2608.10102), submitted.
2. *Meromorphic group actions and the support theorem for Lagrangian fibrations*,
(with M. de Cataldo and C. Schnell), [arXiv:2607.12312](https://arxiv.org/abs/2607.12312), submitted.
3. *Kodaira-type classification of singular fibers of some minimal abelian fibrations*,
[arXiv:2603.02347](https://arxiv.org/abs/2603.02347), submitted.
4. *The Néron model of a higher-dimensional Lagrangian fibration*,
[arXiv:2410.21193](https://arxiv.org/abs/2410.21193), submitted.
5. *Isotrivial Lagrangian fibrations of compact hyper-Kähler manifolds*,
(with R. Laza and O. Martin), J. Math. Pures Appl. (9) **205** (2026), no. 103810.
6. *The dual Lagrangian fibration of known hyper-Kähler manifolds*,
Algebr. Geom. **12** (2025), no. 5.
7. *The LLV decomposition of hyper-Kähler cohomology*,
(with M. Green, R. Laza and C. Robles), Math. Ann. **382** (2022), no. 3–4.
8. *A conjectural bound on the second Betti number for hyper-Kähler manifolds*,
(with R. Laza), Bull. Soc. Math. France **148** (2020), no. 3.

Research talks

Oct 2020	AMS Southeastern meeting	Nov 2020	Stony Brook University
Feb 2021	University of Connecticut	Feb 2021	KIAS, Korea
Sep 2021	Michigan State University	Sep 2021	Center for Complex Geom., Korea
Oct 2021	UC Santa Barbara	Oct 2021	U Massachusetts Amherst
Oct 2021	IMJ-PRG, France	Nov 2021	University of Michigan
Dec 2021	Derived seminar (virtual)	Jan 2022	University of Georgia
Feb 2022	Yale University	Aug 2022	Seoul National University, Korea
Oct 2022	Universität Bonn, Germany	Nov 2022	ISTA, Austria
Feb 2023	Center for Complex Geom., Korea	Feb 2023	KIAS, Korea
Mar 2023	U Amsterdam, Netherlands	Mar 2023	University of Lille, France
Apr 2023	Virtual Symplectic Seminar	Sep 2023	Columbia University
Sep 2023	Stony Brook University	Feb 2024	Center for Complex Geom., Korea
Mar 2024	Princeton University	Mar 2024	Johns Hopkins University
Jun 2024	KIAS, Korea	Sep 2024	Fordham University
Feb 2025	Harvard University – MIT	Feb 2025	University of Sheffield, UK
Feb 2025	University of Pennsylvania	Mar 2025	Center for Complex Geom., Korea
Mar 2025	Yonsei University, Korea	Jun 2025	KIAS, Korea
Sep 2025	Stony Brook University	Nov 2025	IMJ-PRG, France
Nov 2025	Leiden University, Netherlands	Jan 2026	JMM (Fano and K-trivial varieties)
Jan 2026	JMM (Moduli of surfaces)	Feb 2026	Northwestern University
Mar 2026	NCTS, Taiwan	Apr 2026	Princeton University
Apr 2026	University of Georgia	Jun 2026	University of Tokyo, Japan
Jul 2026	KIAS, Korea	Aug 2026	SCMS, China
Sep 2026	CUNY	Oct 2026	University of Chicago
Oct 2026	ICERM (Cubic fourfolds and HK)		

Teaching and Student mentoring

Current teaching

Fall 2026 Linear algebra

Past teaching

Fall 2016	Calculus III (TA)	Spring 2017	Calculus B (TA)
Fall 2017	Calculus C (grader)	Spring 2018	Calculus II (TA)
Fall 2018	Calculus III (TA)	Spring 2019	Linear algebra (TA)
Fall 2019	Precalculus (TA)	Spring 2020	Calculus III (TA)

Fall 2020	Applied algebra (TA)	Spring 2021	Calculus B (TA)
Fall 2021	Calculus III (TA)	Spring 2022	Calculus IV (TA)
(TA means teaching two recitations every week)			
Fall 2023	Calculus II	Spring 2024	Number theory and cryptography
Fall 2024	Calculus III	Spring 2025	Algebraic curves
Fall 2025	Linear algebra	Spring 2026	Algebraic curves

Student mentoring

- Spring 2025 Independent study: Commutative algebra.
- Recommendation letters: 3 undergraduate students.

Services

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| 2024–2025 | Columbia Algebraic Geometry seminar. | <i>Organizer</i> |
| 2025–2026 | Columbia Algebraic Geometry seminar. | <i>Organizer</i> |
| Feb 2025 | FRG derived workshop, Columbia University. | <i>Organizer</i> |
| Mar 2025 | AMS Southeastern sectional meeting, Clemson University. | <i>Organizer</i> |
| | – Columbia AG learning seminars: Spring 24, Spring 25, Spring 26, Fall 26. | <i>Organizer</i> |
| | – 14 referee reports and reviews. | |

Awards and Skills

- Awards
- The presidential science scholarship, 2008–2012, Korea.
 - Silver medal, IMO 2007.
- Language
- English: fluent.
 - Korean: native.
- Programming
- Very fluent in coding with C/C++.

Last edited: September 26, 2026